4 logarytmy 1

Oblicz:

a) $log_9(log_8(log_3 9))$

b) $log_8\left(log_{o,25}(log_4 2)\right)$

Obliczenia logarytmów zaczynamy od tych wewnętrznych

a) $\log_9(\log_8(\log_3 9)) = \log_9(\log_8 2) = \log_9 \frac{1}{3} = -\frac{1}{2}$

$\log_a b = c$ $\qquad a^c = b$

$\log_9 \frac{1}{3} = x$

$9^x = \frac{1}{3}$ $\qquad a^{-n} = \frac{1}{a^n}$

$(3^2)^x = 3^{-1}$

$3^{2x} = 3^{-1}$

$2x = -1 \;|:2$

$x = -\frac{1}{2}$

Odp: $-\frac{1}{2}$.

b) $\log_8(\log_{0,25}(\log_4 2)) = \log_8(\log_{0,25} \frac{1}{2}) =$

$= \log_8 \frac{1}{2} = -\frac{1}{3}$

$\log_{0,25} \frac{1}{2} = y$

$\frac{1}{4}^y = \frac{1}{2}$

$2^{-2y} = 2^{-1}$

$-2y = -1 \;|:(-2)$

$y = \frac{1}{2}$

$\log_8 \frac{1}{2} = z$

$8^z = \frac{1}{2}$

$2^{3z} = 2^{-1}$

$3z = -1 \;|:3$

$z = -\frac{1}{3}$

Odp: $-\frac{1}{3}$.